



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

# ECS PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Cloud

TOTAL: 10 QUESTIONS

**Q1** What is ECS and what is its primary purpose in its cloud ecosystem? Cloud Nat Model

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**Q6** How do you manage configuration and secrets for ECS? ECS Observability

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**Q2** How does IAM work with ECS? Architecture

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**Q7** What are the main cost drivers for ECS and how do you optimize them? Lifecycle

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**Q3** How do you monitor ECS in production? Configuration

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**Q8** How do you implement high availability with ECS? Compliance

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**Q4** How do you scale ECS to handle variable load? Hardening

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**Q9** How does ECS integrate with a CI/CD pipeline? Testing

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**Q5** How do you secure ECS in production? SSO / IdP

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**Q10** What are common pitfalls when first adopting ECS? Tuning

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 ECS is a managed cloud service that

Mitigation

ECS is a managed cloud service that handles a specific infrastructure concern (compute, storage, messaging, AI). Using it shifts operational burden to the cloud provider while you focus on application logic.

A6 Store sensitive values in a managed secret

Observability

Store sensitive values in a managed secret store (AWS Secrets Manager, GCP Secret Manager, Azure Key Vault). Inject secrets as environment variables at runtime, never bake them into container images or source code.

A2 Access to ECS is controlled via IAM

Primitives

Access to ECS is controlled via IAM roles and policies. Attach the minimum required permissions to a service account or role. Avoid long-lived access keys; use instance roles or Workload Identity Federation instead.

A7 Cost in ECS typically comes from compute

Automation

Cost in ECS typically comes from compute hours, data transfer, storage, and request counts. Right-size instances, use reserved capacity for steady-state workloads, and enable lifecycle policies to expire old data.

A3 ECS emits metrics to the cloud provider's

Hardening

ECS emits metrics to the cloud provider's monitoring service (CloudWatch, Cloud Monitoring, Azure Monitor). Set alerts on key metrics (error rate, latency, capacity). Use distributed tracing to correlate across services.

A8 Deploy ECS across multiple availability zones. Use

Governance

Deploy ECS across multiple availability zones. Use managed replicas or active-active configurations. Implement health checks and automatic failover. Test resilience regularly with chaos engineering or failover drills.

A4 ECS supports auto-scaling policies based on CPU,

Gotchas

ECS supports auto-scaling policies based on CPU, queue depth, or custom metrics. Set minimum and maximum capacity. Horizontal scaling (more instances) is generally preferred over vertical for cloud workloads.

A9 CI/CD pipelines use the cloud CLI or

Assessment

CI/CD pipelines use the cloud CLI or SDK to deploy updates to ECS after tests pass. Infrastructure-as-code (Terraform, CDK) defines ECS configuration as version-controlled code deployed via the pipeline.

A5 Place ECS in a private subnet with

Federation

Place ECS in a private subnet with no public endpoint where possible. Use VPC security groups or firewall rules to allow only required traffic. Encrypt data at rest and in transit. Rotate credentials regularly.

A10 Common mistakes include misconfigured IAM policies (too

mitigation

Common mistakes include misconfigured IAM policies (too permissive), no cost alerting, deploying only in one availability zone, and missing backups. Read the service SLA carefully to understand what the cloud provider guarantees.